
Andy Megowan

Software Development Engineer

1133 NE Perkins Way
Shoreline, WA 98155
(206) 850-9992
andymegowan@gmail.com

Senior software engineer with 20+ years of experience delivering resilient, scalable infrastructure and automation across large-scale distributed systems. Proven track record in platform engineering, infrastructure as code (Terraform, CDK, CloudFormation), CI/CD pipeline design, and Python scripting for automation and system orchestration. Led mission-critical automation and platform initiatives at Amazon, AWS, and AI startups. Highly adept at troubleshooting distributed systems, optimizing deployment workflows, and solving deep architectural challenges.

Seeking a contract engineering role focused on platform automation, infrastructure engineering, and DevOps tooling.

CORE COMPETENCIES

Languages: Python, Bash, Rust, Go, Java

Infrastructure as Code: Terraform, CloudFormation

Automation & DevOps: CI/CD, Jenkins, GitHub Actions, CloudWatch

Containers & Orchestration: Docker, Kubernetes, ECS, EKS

Cloud Platforms: AWS (EC2, Lambda, S3, IAM, VPC, SNS/SQS)

Monitoring: CloudWatch, Splunk, Rollbar, NewRelic, Nagios

Systems: Linux, Shell Scripting, Load Balancing, Networking

Microservices: REST APIs, Rails, Flask, Dropwizard, Golang

Databases: PostgreSQL, Redis, MySQL, DynamoDB, MongoDB (admin & schema)

Development Practices: Agile, Scrum, Git, TDD, Configuration Management

Frameworks: Coral/Swagger, Flask, FastAPI, pytest, boto3

AI / Automation: LLMs, Prompt Engineering, AWS AI/ML (SageMaker, Lambda), TTS Pipelines

CI/CD: Jenkins, GitHub Actions, Terraform, Docker, Kubernetes

Tools: Docker, Kubernetes, Terraform, Git, Jenkins, Flask, Coral

Cloud: AWS (S3, EC2, Lambda, RDS, CloudFormation, CloudWatch), GCP (familiar)

Databases: PostgreSQL, Redis, DynamoDB, Aurora

Source Control / Tools: Git, Perforce, Jira, Confluence, VSS, Trello

AI Coding assistants: Cursor, Copilot, ChatGPT, Claude, Windsurf

EXPERIENCE

[BitPart.ai](#) - *Software Engineer - AI Platform (Contract, Remote)*

November 2024 - April 2025

- Developed infrastructure to support AI-based audio generation and delivery using Terraform, Python, Rust, TypeScript and cloud-native tooling (S3, Lambda) with GitHub Actions for CI/CD.
- Integrated media services into cross-platform clients (Unity, Unreal)
- Enhanced system throughput and stability by profiling async workloads and restructuring the SQLAlchemy interactions..

Amazon - Compute Technologies - *SDE II, SysDE II Platform & Infrastructure Engineering*

April 2022 - June 2024

- Architected and maintained scalable infrastructure using CDK and CloudFormation to manage 200K+ services across AWS.
- Led design and rollout of enterprise CI/CD pipelines, including deployment automation, secure secrets rotation, and configuration management.
- Developed automation scripts in Python and Bash to generate system dashboards, monitor traffic proxies, and detect drift across hosts.
- Supported Dockerized services and managed Kubernetes-like clusters for internal tooling and network services.
- Diagnosed distributed failures across AWS-hosted microservices and improved system observability with CloudWatch.

AWS - Real-Time Protocol - *SDE II - Streaming Platform Engineer*

January 2020 - March 2022

- Designed metric ingestion and publishing systems using Python, C++, and CloudFormation to analyze real-time performance of distributed protocols.
- Built automation to triage protocol and regression failures using custom scripts and alerts.
- Managed multiple staging environments with Docker-based orchestration; reduced deploy-to-test latency by 45%.

Amazon Mechanical Turk - *SDE II*

June 2017 - December 2019

- Designed and integrated the MTurk fraud detection system.
- (As Manager) Ran the Operational Readiness for 2019 launch of AWS Augmented Artificial Intelligence (A2I)

-
- Created the new Worker Qualification pipeline to allow selection of workers by demographic.
 - Implemented SSO (Single Sign-On) for the old and new versions of the MTurk website.

zulily, Seattle - *Software Engineer (Supply Chain, Transportation, Store)*

November 2013 - June 2017

- Received very first Zulily Employee Award for my work on fraud prevention, which saved the company an estimated \$3.1M in the first year.
- Allow customers to seamlessly move between web store and mobile app.
- Put all of Tech on Database APIs instead of direct database queries.
- Overhauled purchase logic to allow payment in installments.
- Built the microservices for: customer search and autocomplete, free shipping logic, split payments, global sales event availability, fraud detection

zSpace, Inc, Seattle- *Lead Developer, Studio Manager*

June 2011 - February 2013

- Opened and ran a 10-person, 2-location studio for a California business.
- Acted as designer, project manager, and lead programmer on all projects.
- Developed games and applications to explore interactivity with mixed reality for the ZSpace hardware.

EDUCATION

University of California, Berkeley

Bachelor of Architecture (with Computer Science specialization)

Certification & Continuing Education:

- AWS Cloud Practitioner
- Prompt Engineering (Coursera)

RECENT PROJECTS

- **Voice Pipe (2025):** Cloud-based orchestration of AI-driven pipeline: text, translation, speech, lip-sync.
- **CorrectiveLens (2025):** Felon-focused device monitoring for employers, therapists, and law enforcement.